Reflections

Exam code: 0607

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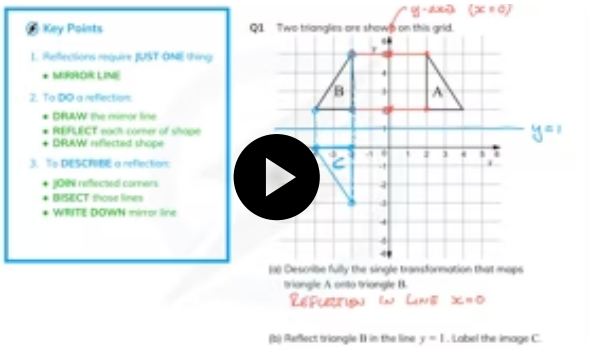

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Help me understand

Reflections

What is a reflection?

- A **reflection flips** a shape across a **mirror line**
  - This is called the **line of reflection**
- The **reflected image** is the **same size** as the **original object**
  - It has been **flipped** across the mirror line to a **new position** and **orientation**
- The following two **distances will be equal** for each point:
  - The **perpendicular distance** between the **original point and the mirror line**
  - The **perpendicular distance** between the **reflected point and the mirror line**
- Any **points** that are **on the mirror line** do not move
  - These are called **invariant points**

How do I reflect a shape?

- **STEP 1**
  Draw the **line of reflection**
  - This will usually be a **vertical** line ($x = k$) or a **horizontal** line ($y = k$)
  - A **diagonal** line will either be $y = x$ or $y = -x$

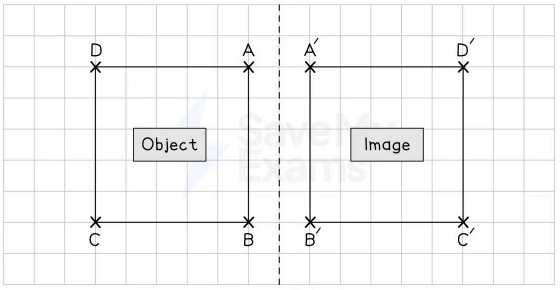
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- You can usually do this by **counting squares** on the grid
- If the line is **diagonal** then count the **diagonals of the squares**

- **STEP 3**
  Find the **reflected** point by **measuring** the **same distance** in the **same direction** from the point on the mirror line
- **STEP 4**
  Join together the reflected points and label the reflected image



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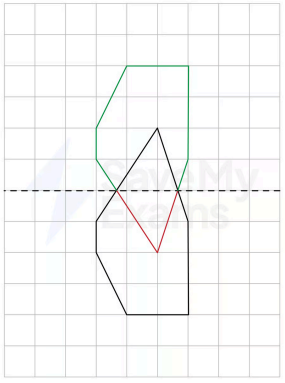
How do I reflect a shape when the line of reflection goes through the shape?

- You follow the **same steps** as above
- Part of the shape gets reflected on **one side** of the mirror line, and the other part gets reflected on the other side



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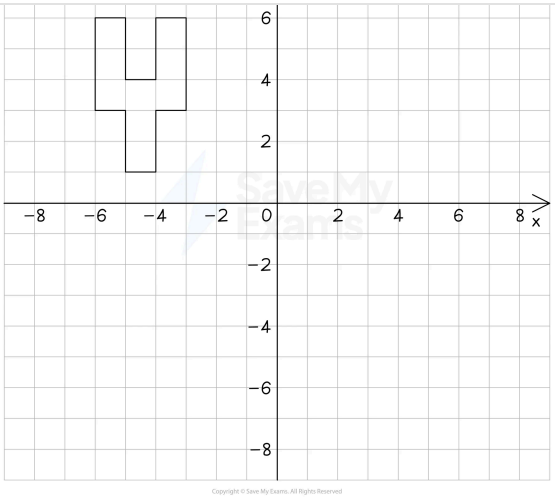
Worked Example

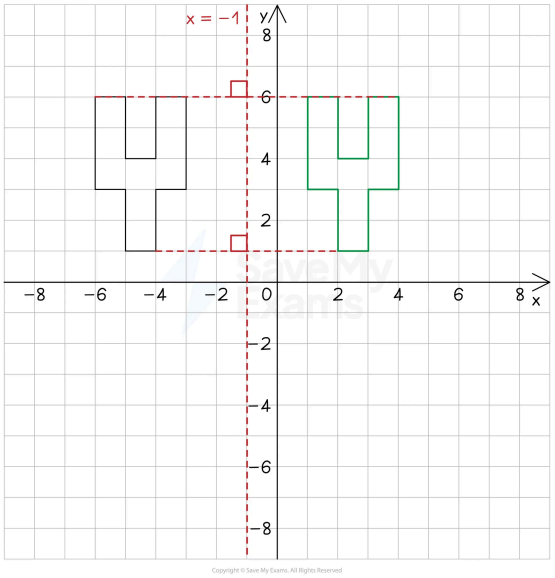
On the grid below, reflect shape S in the line $x = -1$.

State the coordinates of all of the vertices of your reflected shape.



Answer:

Draw in the mirror line; $x = -1$ will be a vertical line passing through -1 on the x-axis
Measure or count the number of units from the shape "diagonals" on the other side of the mirror line to find the position of the corresponding vertex on the reflected image



List the vertices of the reflected image.
Work your way around the shape vertex by vertex so that you don't miss any out as there are quite a few!

Vertices of the reflected shape: (1, 6), (2, 6), (2, 4), (3, 4), (3, 6), (4, 6), (4, 3), (3, 3), (3, 1), (2, 1), (2, 3), (1,3)

How do I describe a reflection?

- To describe a **reflection**, you must:
  - State that the transformation is a **reflection**
  - Give the mathematical **equation of the mirror line**

the y-axis

- **Vertical** lines are of the form $x = k$
  - k is the number that the **line passes through on the x-axis**

- A **diagonal** line with a **positive gradient** will be $y = x$
- A **diagonal** line with a **negative gradient** will be $y = -x$

How do I reverse a reflection?

- If a shape has been reflected to a new position, you can perform a single transformation to **return the shape** to its **original position**
  - You can **reverse the reflection**

- The transformation to **reverse a reflection**, is the **same transformation** as the **original**
  - E.g. If a shape is reflected in the *x*-axis, then reflecting it again in the *x*-axis will return it to its original position

Examiner Tips and Tricks

It is very easy to muddle up the equations for **horizontal** and **vertical** lines, remember:

- If the line crosses the **x-axis** then it will be $x = k$
- If the line crosses the **y-axis** then it will be $y = k$

Worked Example

Describe fully the single transformation that creates shape B from shape A.

without too much difficulty.
Draw the mirror line on the diagram.
You can check that it is in the correct position by measuring/counting the perpendicular distance from a pair of corresponding points on the original object and the reflected image to the same point on the mirror line.
Be careful with mirror lines near axes as it is easy to miscount.

Write down that the transformation was a reflection and the equation of the mirror line.

Shape A has been reflected in the line $x = -1$ to create shape B

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